TSM 098 Final Exam

Practice 2 | Version 202610122020.14

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Solve and check. $3x - 3 = -x + 6 + 2x$	1. Solution 10 pts
2. Evaluate at x = -4, y = 2. Show substitution. $3x - 3y + 1$	2. Value 10 pts
3. Add and combine like terms. $(3x^2 + 3x - 4) + (-x^2 - 5x + 2)$	3. Polynomial 10 pts
4. Multiply and simplify. $(4x - 4)(x + 4)$	4. Polynomial 10 pts
5. Distribute and simplify. $-3x^2(2x^2 - 4x + 4)$	5. Polynomial 10 pts
6. Add and reduce to lowest terms. $\frac{2}{9} + \frac{1}{6}$	6. Fraction 10 pts
7. Solve by substitution or elimination; check both equations. $y = 2x - 6$ $x + y = 6$	7. Ordered pair 10 pts
8. Subtract; show distribution of the negative and combine like terms. $(2x^2 - 2x + 3) - (x^2 + 3x - 4)$	8. Polynomial 10 pts
9. Solve and state the complete solution set. $3(x + 4) - 2 = 3x + 10$	9. Solution set 10 pts
10. Solve and check in the original equation. $\frac{x+4}{3} + \frac{x}{4} = 3$	10. Solution 10 pts

Instructor answer key: TSM 098 Final Exam

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. Solution

$x = \frac{9}{2}$

10 pts

2. Value

-17

10 pts

3. Polynomial

$2x^2 - 2x - 2$

10 pts

4. Polynomial

$4x^2 + 12x - 16$

10 pts

5. Polynomial

$-6x^4 + 12x^3 - 12x^2$

10 pts

6. Fraction

$\frac{7}{18}$

10 pts

7. Ordered pair

$(4,\ 2)$

10 pts

8. Polynomial

$x^2 - 5x + 7$

10 pts

9. Solution set

All real numbers; both sides agree after distribution.

10 pts

10. Solution

$x = \frac{20}{7}$

10 pts